

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 25/JAN 26

SUBJECT: CSS 2104/CSS_2104 DIGITAL SYSTEMS COMPUTER ORGANIZATION

(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) In a signal processing application, two equal-length vectors A and B represent input and weight values of a digital filter. Each vector contains N elements, and the dot product of these vectors determines the filter output. Develop a CISC-style assembly program to compute the dot product of vectors A and B and store the result in memory location DOTPROD. Describe the sequence of operations required through suitable codes and explanations. (4)

Course Outcome: "CO5" / Cognitive Level: "Create"

- 1B) Represent the following numbers in 5-bit 2's complement form, perform the operation mentioned and state whether overflow occurs or not in each case. (3)
- i) Add +9 and +12
 - ii) Add -8 and -9
- Subtract -5 from +10

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If E = 0, the circuit remains in the same state regardless of the value of F. When E = 1 and F = 1, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When E = 1 and F = 0, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (5)

Course Outcome: "CO2" / Cognitive Level: "Create"

- 2B) Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. (3)

Course Outcome: "CO2" / Cognitive Level: "Create"

- 2C) Write the Verilog code for the K flip flop whose function Table is given below (2)

Clock	K	Output
0	X	No Change
1	0	1
1	1	0

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 3A) A block-set-associative cache consists of a total of 128 blocks, divided into 8-block sets. The main memory contains 8192 blocks, each consisting of 256 words. (4)
- i) How many bits are there in the main memory address?
- ii) How many bits are there in each of the TAG, SET, and WORD fields?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) i) Write the sequence of Register Transfer Notations and Assembly language statements to compute (3)

$$X = (A + B) - (C + D)$$

Assume each variable represents a memory location.

- ii) Explain the role of the Program Counter (PC) and Instruction Register (IR) during instruction sequencing.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 3C) Given $x=0101$ and $y=1010$, (3)
- Compute the product $p=x*y$ using Booth's algorithm. Show all the steps involved.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) i) Distinguish between Uniform Memory Access and Non Uniform Memory Access multi processors. (4)
- ii) Explain how cache coherence is maintained in Write-Through and Write-Back protocols.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 4B) Perform the addition of the following two 32-bit IEEE 754 single-precision floating-point numbers (3)
- and represent the result in IEEE 754 format.
- i) $A = 0 \ 10000010 \ 1010000000000000000000$
- ii) $B = 0 \ 10000001 \ 1100000000000000000000$
- Show all the steps including conversion to decimal, exponent alignment, addition, normalization, and final binary representation.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) Apply the concept of microprogramming to explain how a control unit generates control signals during instruction execution, using a suitable diagram. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

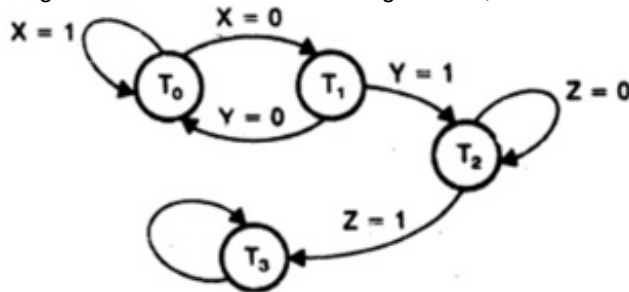
- 5A) Draw the complete processing section diagram for the following register transfer description indicating all the control points, inputs and outputs. List out the control points and the operation performed. (4)

Declare registers A[4], M[4], Q[4], L[3], C[1]
 Declare buses inbus[4], outbus[4]
 Start: $A \leftarrow 0, C \leftarrow 0, M \leftarrow \text{inbus}, L \leftarrow 4;$

$Q \leftarrow \text{inbus}$
 Loop: if $Q[0] = 1$, then go to ADD
 Go to Rshift;
 ADD: $A \leftarrow A + M;$
 Rshift: $\text{LSR}(C\$A\$Q), L \leftarrow L - 1;$
 if $L > 0$, then go to Loop
 outbus $\leftarrow A;$
 outbus $\leftarrow Q;$
 Halt: Go to Halt

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5B) A control circuit has three inputs X, Y and Z and its state diagram is shown below, (3)
 Design the hardwired controller using counter, decoder and sequence controller.



Course Outcome: "CO3" / Cognitive Level: "Application"

- 5C) Implement the 4×4 restoring integer division algorithm using appropriate register transfer (3)
 operations.

Course Outcome: "CO3" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSS_2104:Digital Systems and Computer Organization

#	Course Outcome	Description	Marks
1	1	CO 1	6
2	2	CO 2	10
3	3	CO 3	19
4	4	CO 4	8
5	5	CO 5	7
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSS_2104:Digital Systems and Computer Organization

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	6	18(36%)
4	Analysis	4	14(28%)
5	Create	2	6(12%)
6	Evaluate	3	12(24%)
	Total	15	50 (100%)